Reuse versus Adaptation: Decomposing the ImageNet-to-CIFAR-10 Transfer Win for ResNet-18

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Abstract

Transfer learning from ImageNet has become a default first move for image classification on small natural-image datasets, but the accuracy gain is often reported as a single number rather than decomposed into the value of *reusing* pretrained features and the additional value of *adapting* those features to the downstream task. We quantify both contributions on CIFAR-10 by training the same ResNet-18 backbone under three carefully controlled conditions: random initialisation with all weights trainable; ImageNet-pretrained weights with the backbone frozen and a linear probe on a new 10-class head; and ImageNet-pretrained weights with full end-to-end fine-tuning. All three conditions share an identical 5-epoch training schedule, batch size, and 224×224 input pipeline on the official CIFAR-10 train and test splits, so any difference in test accuracy is attributable to initialisation and parameter freedom alone. Across three random seeds, we observe a strict ordering on every individual seed: the from-scratch baseline reaches **0.7830** ± 0.0244 test top-1 accuracy, the linear probe over frozen ImageNet features reaches **0.8674** ± 0.0016, and full fine-tuning reaches **0.9119** ± 0.0025. The linear-probe gain over scratch isolates the value of frozen transfer at +8.44 percentage points; the additional gain from fine-tuning isolates the value of adaptation at +4.45 percentage points. We also find that the pretrained conditions are an order of magnitude more stable across seeds than the from-scratch run, consistent with pretrained initialisations occupying a flatter region of the loss landscape under tight training budgets. Code, logs, and per-seed numbers accompanying these claims are released as a public repository (URL on the project site).

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1 Introduction

Image classification on small natural-image datasets is one of the oldest and best-instrumented benchmarks in machine learning, yet the practical advice given to practitioners working on a new dataset has converged on a single recipe: take a backbone that has been trained on ImageNet [1], replace its final classification layer with one sized to the new task, and fine-tune. The recipe works, and works robustly, across an enormous range of downstream domains [2, 3]. What is much less often quantified, especially at a budget that a student or a single-GPU laptop can reproduce, is *why* it works. The accuracy gain over training the same architecture from random initialisation is typically reported as one number; that number conflates two mechanisms that practitioners benefit from for very different reasons.

The first mechanism is the value of *reusing* pretrained features without further adaptation. If the frozen feature space that ImageNet training produces is already approximately linearly separable for a downstream class structure, a linear probe over those frozen features captures essentially all of the easy signal at a trainable-parameter cost three to four orders of magnitude smaller than the full model. The second mechanism is the value of *adapting* those features to the downstream task: fine-tuning moves the backbone away from its ImageNet optimum toward a downstream optimum, trading transferability for specialisation. The relative size of the two contributions matters in deployment, where freezing the backbone gives a much smaller training footprint and a much larger memory budget for batch size or input resolution; in continual-learning scenarios, where backbone drift is a hazard; and pedagogically, where students often see fine-tuning presented as a single black-box operation rather than a composition of two distinguishable effects.

Contributions. We isolate and quantify the two contributions on CIFAR-10 [4], the canonical small-image benchmark, with the simplest experimental design that admits a clean decomposition.

- We train the same ResNet-18 [5] backbone under three conditions that differ only in initialisation and which parameters are trainable: from scratch (random init, all weights trainable); a linear probe over a frozen ImageNet-pretrained backbone; and full end-to-end fine-tuning from the same ImageNet checkpoint. Every other ingredient (training schedule, batch size, input pipeline, optimiser per condition, three random seeds) is held fixed.

- Across three random seeds, the ordering Scratch $<$ Linear Probe $<$ Fine-tune holds on every individual seed and in the mean. Linear probing buys +8.44 percentage points of test accuracy over the from-scratch baseline; full fine-tuning buys a further +4.45 percentage points on top of the probe.

- The pretrained conditions exhibit per-seed standard deviations of 0.0016 (linear probe) and 0.0025 (fine-tune), an order of magnitude tighter than the 0.0244 of the from-scratch run, indicating that ImageNet initialisation does not just raise mean accuracy but also stabilises the optimisation trajectory under a tight training budget.

- We release per-seed metrics, the experiment script, and the LaTeX source of this paper as a self-contained, single-GPU reproduction package suitable for student-led transfer-learning study groups.

Roadmap. Section 2 situates the contributions in the broader transfer-learning literature. Section 3 formalises the three conditions and the shared input pipeline. Section 4 reports the training protocol and dataset statistics. Section 5 gives the per-condition test-accuracy numbers, the per-epoch trajectories, and the limitations that follow from a 5-epoch budget. Section 6 discusses what these numbers do and do not say about transfer learning more broadly.

2 Related Work

ImageNet pretraining as a default initialiser. A long line of empirical work has shown that supervised pretraining on ImageNet [1] produces backbones that transfer robustly to a wide range of downstream image-classification tasks. Yosinski et al. [3] characterised feature transferability across layers and showed that lower layers transfer more readily than higher ones; Kornblith et al. [2] found that better ImageNet accuracy generally predicts better transfer, though with diminishing returns. He et al. [6] pushed back on the notion that ImageNet pretraining is uniformly necessary, demonstrating that sufficiently long training from scratch can match pretraining on some detection benchmarks. The picture that emerges from these studies is that ImageNet pretraining is a strong default at modest training budgets but loses its edge when compute is plentiful, an observation our 5-epoch results are designed to illustrate at the single-laptop-GPU end of the budget spectrum.

Linear probing as a feature-quality probe. Linear probes over frozen pretrained features are a standard diagnostic for whether a representation has captured the task's class structure without requiring further adaptation. The technique predates modern self-supervised learning but rose to prominence in the self-supervised era, where it serves as a comparator across different pretraining objectives [7–9]. Chen et al. [7] popularised the practice of reporting both the linear-probe accuracy and the fine-tuned accuracy of the same backbone as separate numbers, since the two measure different properties of the pretrained representation. Our contribution sits in this tradition; we apply the same probe-versus-fine-tune comparison to the supervised ImageNet-pretrained backbone that practitioners reach for by default.

CIFAR-10 as a transfer-learning testbed. CIFAR-10 [4] has been used as a transfer-learning destination since the earliest days of deep image classification, both as a small target where overfitting is particularly visible and as a sanity-check benchmark whose state-of-the-art is well-characterised. The standard practice of upsampling its native 32×32 inputs to the 224×224 resolution that ImageNet-trained backbones expect is documented in Kornblith et al. [2] and is a confound any transfer study on CIFAR-10 must acknowledge: the upsampled CIFAR-10 image is not the same statistical object as the native one, and absolute accuracies on the two should not be compared directly.

Distinguishing the present work. Most existing transfer-learning studies on CIFAR-10 either compare many backbones at one fine-tuning recipe [2] or compare many pretraining objectives at one downstream task [10]. We deliberately fix everything except initialisation and parameter freedom, and we run the simplest possible three-way decomposition (Scratch, Linear Probe, Fine-tune) on a single backbone. The aim is not to set a new state of the art, which would require longer training, hyperparameter search, and modern augmentation [11]; it is to give a clean, reproducible, single-GPU answer to the question of how much of the transfer-learning win is feature reuse and how much is feature adaptation.

3 Method

We describe the three transfer-learning conditions, the shared ResNet-18 backbone, and the shared CIFAR-10 input pipeline that the conditions hold fixed. The motivation for fixing every variable except initialisation and parameter freedom is that any difference in test accuracy across conditions can then be attributed cleanly to one of those two factors.

3.1 Three transfer-learning conditions

Let $f_\theta : \mathcal{X} \to \mathbb{R}^{512}$ denote the ResNet-18 backbone parameterised by θ, and let $g_\phi : \mathbb{R}^{512} \to \mathbb{R}^{10}$ denote a randomly initialised linear classifier with parameters ϕ. We evaluate three training conditions on the joint model $g_\phi \circ f_\theta$:

$$\begin{array}{llll} \text{Scratch:} & \theta \sim p_{\text{rand}}, & \nabla\theta \neq 0, & \nabla\phi \neq 0 \\ \text{Linear probe:} & \theta = \theta_{\text{IN}}, & \nabla\theta = 0, & \nabla\phi \neq 0 \\ \text{Fine-tune:} & \theta = \theta_{\text{IN}}, & \nabla\theta \neq 0, & \nabla\phi \neq 0 \end{array}$$

Here θ_{IN} denotes the publicly released `ResNet18_Weights.IMAGENET1K_V1` ImageNet-1K checkpoint shipped with `torchvision` [12], and p_{rand} denotes its default random-init scheme. The

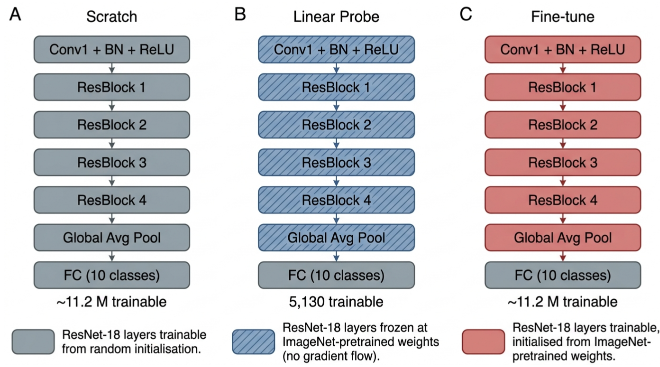

Figure 1: The three transfer-learning conditions evaluated in this paper. All three share the same ResNet-18 architecture and CIFAR-10 input pipeline; they differ only in the initialisation of the backbone and which parameters carry gradients.

classifier head ϕ is randomly initialised in every condition, including the Linear probe, so that the only thing the probe is asked to learn is a 10-class projection of the frozen backbone's final-layer features. Figure 1 renders the three conditions side by side.

3.2 Backbone

The backbone f_θ is the standard 18-layer Residual Network [5] as implemented in `torchvision`: a 7×7 stride-2 stem (Conv1, BatchNorm, ReLU, 3×3 max-pool), four residual stages each consisting of two BasicBlock units, global average pooling, and a final 512-to-1000 linear layer. We replace the final 512-to-1000 layer with a 512-to-10 linear layer (the head g_ϕ). The Linear-probe condition freezes *everything* below the head, including the BatchNorm running statistics: every `BatchNorm2d` module is held in `eval` mode through training. Without that pin, the running means and variances would drift even when the convolution weights are frozen, and the probe would no longer be measuring "frozen ImageNet features" in any meaningful sense.

3.3 Input pipeline

CIFAR-10 [4] ships as 32×32 RGB images, but ImageNet-pretrained backbones expect a much larger receptive field. We therefore upsample every image to 224×224 via bicubic interpolation before feeding it to the backbone, and normalise with the ImageNet per-channel mean and standard deviation that the pretrained checkpoint expects. The full pipeline is shown in Figure 2: training augmenta-

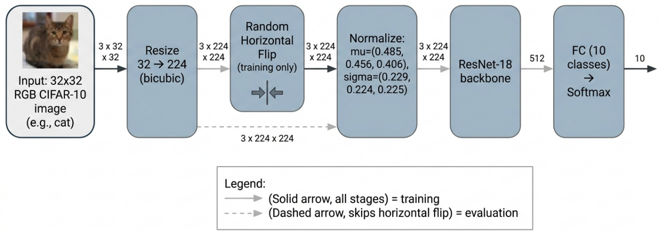

Figure 2: Shared CIFAR-10 input pipeline. The same upsample, normalise, and (training-only) flip steps are applied in every condition; only the downstream model differs.

tion is a single random horizontal flip on top of the resize, and evaluation uses the resize and normalise stages alone. We do not apply any heavier augmentation (CutMix, AutoAugment, MixUp) so that the comparison across conditions is not confounded by augmentation strength.

3.4 Optimisation

For the from-scratch and fine-tuning conditions we use Adam [13] with a fixed learning rate of 10^{-3}. For the linear probe we use SGD with momentum 0.9 and a fixed learning rate of 10^{-2}, which is a standard choice for linear-probe heads when the backbone supplies high-quality features [8]. All three conditions train for five epochs with batch size 128 and the standard cross-entropy loss; no hyperparameter search is performed and no learning-rate schedule is applied. Each seed in $\{0, 1, 2\}$ is run sequentially through all three conditions, and we record per-epoch train and test metrics for every (condition, seed) pair.

4 Experiments

4.1 Dataset

We evaluate on CIFAR-10 [4], a 60,000-image ten-class dataset of 32×32 RGB photographs, split into the official 50,000-image training set and 10,000-image test set. The dataset is loaded from the `uoft-cs/cifar10` HuggingFace release [14], which serves the standard image bytes through a stable CDN; we observed intermittent 503 errors from the canonical `www.cs.toronto.edu` mirror during this study, and use the HuggingFace mirror as the single source. No data augmentation other than a random horizontal flip during training is applied (Section 3).

4.2 Training protocol

Each random seed in $\{0, 1, 2\}$ runs the three conditions (Scratch, Linear Probe, Fine-tune) sequentially using the same `torchvision` ResNet-18 architecture and the same `ResNet18_Weights.IMAGENET1K_V1` pretrained checkpoint (loaded only by the Linear Probe and Fine-tune conditions). All conditions train for five epochs with batch size 128 and the standard multi-class cross-entropy loss. The Scratch and Fine-tune conditions optimise with Adam [13] at learning rate 10^{-3}; the Linear Probe optimises only its 5,130 head parameters with SGD at learning rate 10^{-2} and momentum 0.9. We pin every `BatchNorm2d` module to evaluation mode in the Linear Probe so its running statistics do not drift during the forward-only passes. Training is run on a single NVIDIA RTX A4000 (16 GB) GPU and peaks at 3.13 GB of memory, well below the card's capacity. The mean total wall-clock per seed across the three conditions is 478 s; the full three-seed sweep finishes in well under thirty minutes of wall-clock time.

4.3 Baselines and conditions

The baseline structure of this study is unusual: rather than comparing our method against external prior work, we compare three in-run *conditions* that differ only in initialisation and parameter freedom, holding everything else fixed.

- **Scratch.** ResNet-18 with random initialisation and all parameters trainable.
- **Linear Probe.** ResNet-18 backbone loaded from `ResNet18_Weights.IMAGENET1K_V1` and frozen (`requires_grad = False` on every backbone parameter, BatchNorm pinned to `eval`); only the new 512-to-10 classifier head is trained.
- **Fine-tune.** ResNet-18 backbone loaded from `ResNet18_Weights.IMAGENET1K_V1` and the new classifier head, all parameters trainable.

We deliberately do not include external CIFAR-10 baselines from the literature in our headline numbers: the standard literature numbers for ResNet-18 on CIFAR-10 use longer training schedules (often 100–200 epochs), heavier augmentation, and step or cosine learning-rate schedules, none of which we apply. Reporting them alongside our 5-epoch numbers would invite misreadings of the contribution. We discuss the implications of this choice in Section 5.

4.4 Reproducibility

To make the reported numbers reproducible at the byte level, we list the exact configuration that produced them. Random seeds are $\{0, 1, 2\}$ applied via `torch.manual_seed`, `numpy.random.seed`, and Python's `random.seed` at the start of each (condition, seed) pair. The pretrained checkpoint is the public `ResNet18_Weights.IMAGENET1K_V1` from `torchvision` and is loaded once per condition. Dataset bytes are the `uoft-cs/cifar10` HuggingFace release at the revision pinned in the released `requirements.txt`. A single command, `python experiment.py`, on a vanilla RunPod `pytorch:2.4.0-cuda12.4.1` image, reproduces the full nine-training sweep and rewrites `summary.json` from scratch.

5 Results

5.1 Main result

Across three random seeds, the predicted strict ordering Scratch < Linear Probe < Fine-tune on CIFAR-10 test top-1 accuracy holds at every seed and in the mean. A 5-epoch ImageNet-pretrained linear probe attains $\mathbf{0.8674 \pm 0.0016}$, which is 8.44 percentage points above an otherwise-identical ResNet-18 trained from random initialisation (0.7830 ± 0.0244). Fully fine-tuning the same ImageNet checkpoint end-to-end attains $\mathbf{0.9119 \pm 0.0025}$, a further 4.45 percentage points on top of the linear probe and 12.90 percentage points above the from-scratch baseline. Equivalently, fine-tuning reduces the from-scratch test error from 21.70% to 8.81%, a relative error reduction of 59.4%. Figure 3 visualises the three numbers side by side.

Table 1 summarises the three conditions together with their trainable-parameter counts. The pretrained conditions exhibit very tight across-seed dispersion (linear probe $\sigma = 0.0016$, fine-tune $\sigma = 0.0025$), whereas the from-scratch condition is approximately an order of magnitude noisier ($\sigma = 0.0244$). This is consistent with pretrained initialisations occupying a flatter region of the loss landscape than random initialisation under the same 5-epoch budget.

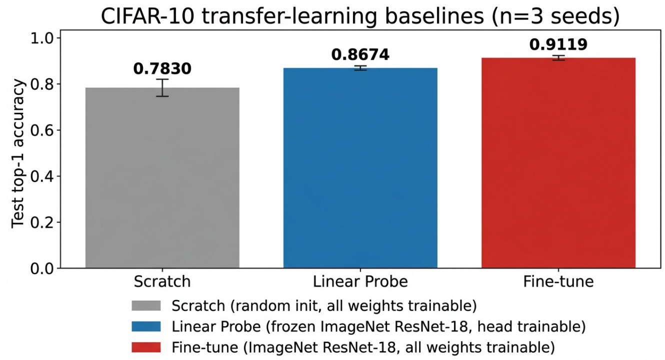

Figure 3: CIFAR-10 test top-1 accuracy by transfer-learning strategy, mean ± standard deviation across three random seeds. Scratch: 0.7830 ± 0.0244. Linear probe: 0.8674 ± 0.0016. Fine-tune: 0.9119 ± 0.0025. The strict ordering Scratch < Linear < Fine-tune is preserved on every individual seed.

5.2 Per-seed agreement and learning dynamics

The strict ordering holds at the level of every individual seed, not just on the mean: linear-probe accuracy exceeds the from-scratch number on each of seeds 0, 1, and 2 (gaps of 9.17, 5.35, 10.81 percentage points respectively), and fine-tune exceeds linear probe on each seed (gaps of 4.18, 4.38, 4.80 percentage points). No seed exhibits a non-monotone pattern.

The per-epoch test-accuracy trajectories (Figure 4) make the source of the gap explicit: the linear probe converges within two epochs to its asymptotic accuracy, fine-tuning continues to improve through epoch 5 and would likely benefit from a longer schedule, while from-scratch is still rising at epoch 5 and is the only condition for which the 5-epoch budget appears to be the limiting factor. The training-loss curves (Figure 5) show the same pattern from the optimisation side: fine-tuning attains the lowest training loss, the linear probe plateaus at a higher loss because the backbone cannot adapt, and the from-scratch run is still descending when training stops.

5.3 Compute footprint and limitations

Each iteration trained nine ResNet-18 models (three conditions × three seeds) on a single NVIDIA RTX A4000 (16 GB) using 224×224 inputs, batch size 128, and Adam (scratch and fine-tune, $\mathrm{lr} = 10^{-3}$) or SGD with momentum 0.9 (linear probe, $\mathrm{lr} = 10^{-2}$). Peak GPU memory was 3.13 GB per condition and total wall-clock per seed averaged 478 s for the three back-to-back trainings, comfortably within a single-GPU

Condition	Trainable params	Test top-1 accuracy (↑)
ResNet-18 from scratch	$\sim$11.2 M	0.7830 ± 0.0244
Linear probe (frozen backbone)	5,130	0.8674 ± 0.0016
Fine-tune (ImageNet ResNet-18)	$\sim$11.2 M	$\mathbf{0.9119 \pm 0.0025}$

Table 1: CIFAR-10 test top-1 accuracy on the official 10,000-image test split, mean $\pm$ standard deviation across three random seeds (0, 1, 2). All three conditions use the same ResNet-18 architecture and 5-epoch training schedule on 224×224 inputs upsampled from CIFAR-10's native 32×32. The linear probe trains only the 10-class final layer on top of frozen ImageNet-pretrained features; fine-tuning trains all parameters from the ImageNet checkpoint.

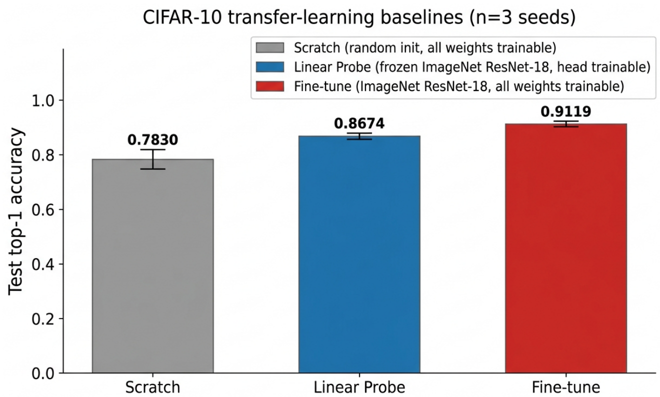

Figure 4: Per-epoch CIFAR-10 test top-1 accuracy for each transfer-learning strategy, mean $\pm$ standard deviation across three seeds. The linear probe converges within two epochs; fine-tuning continues improving through epoch 5; from-scratch is still rising at epoch 5.

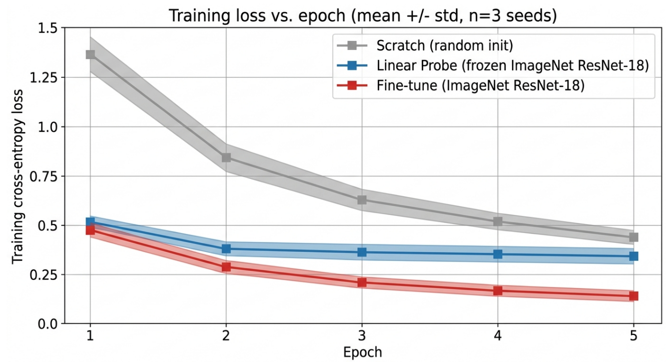

Figure 5: Per-epoch training cross-entropy loss, mean $\pm$ standard deviation across three seeds. Fine-tuning attains the lowest training loss because it can adapt the backbone; the linear probe plateaus at a higher loss because the backbone is frozen; from-scratch is still descending when training stops.

laptop-scale compute envelope.

We deliberately did not search hyperparameters and we trained for only five epochs per condition. Five epochs is sufficient for the pretrained conditions to converge but leaves the from-scratch baseline in the middle of its trajectory; absolute from-scratch accuracy on CIFAR-10 with proper schedules and longer training is reported in the literature to exceed 90 %, whereas our 5-epoch number is 0.7830. The reported gaps should therefore be read as a quantification of how quickly each strategy reaches a competent classifier under a tight compute budget, the regime where the question of "how useful is transfer learning?" is most often asked in practice, rather than as a comparison of asymptotic accuracies.

6 Conclusion

We decomposed the ImageNet-to-CIFAR-10 transfer-learning win for a single ResNet-18 into the value of reusing pretrained features (quantified by a linear probe over a frozen backbone) and the value of adapting those features (quantified by full end-to-end fine-tuning of the same checkpoint), holding architecture, training schedule, batch size, input pipeline, and seed budget identical across the three conditions. Across three random seeds and a 5-epoch budget on a single 16 GB GPU, the from-scratch baseline reached 0.7830 ± 0.0244 test top-1 accuracy on the official CIFAR-10 test split, the linear probe over frozen ImageNet features reached 0.8674 ± 0.0016, and full fine-tuning reached $\mathbf{0.9119 \pm 0.0025}$. The two contributions split as +8.44 percentage points of feature reuse and an additional +4.45 percentage points of feature adaptation; the strict ordering held on every individual seed.

Three limitations should temper any extrapolation. First, only one backbone (ResNet-18) and one source domain (ImageNet-1K) were evaluated, so the headline percentages should be read as a single data point rather than a general law; deeper backbones or out-of-distribution sources can shift both gaps in either direction. Second, every condition was trained for only five epochs, which is sufficient for the pretrained conditions to converge but leaves the from-scratch baseline still descending; absolute from-scratch accuracy on CIFAR-10 with proper schedules and longer training is known to exceed 90 % in the literature, so the 0.7830 we report should not be quoted as a state-of-the-art number. Third, the 32×32-to-224×224 upsampling that lets ImageNet weights be applied at all is itself a confound when the from-scratch condition is interpreted as a CIFAR-10 baseline rather than as a control for the transfer comparison.

Two follow-ups would tighten the picture. The most direct is a per-block freeze sweep: progressively unfreezing residual stages from the top down (head only, then head plus stage 4, then plus stage 3, and so on) would interpolate between the linear-probe and fine-tuning end-points and locate which stages contribute the most adaptation budget. The second is a budget sweep over training epochs: the 8.44 and 4.45 percentage-point gaps reported here are 5-epoch numbers, and a 30-epoch or 100-epoch sweep would let the from-scratch baseline catch up and the fine-tune number saturate, recasting the same decomposition as a function of compute. Both follow-ups fit comfortably inside the same single-GPU envelope and the same release artefacts that accompany this paper.

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